

Partial Fractions



Distinct linear factors

- 1 Express $\frac{5x+7}{(x+1)(x+2)}$ in partial fractions.
 - 2 Express $\frac{3x-1}{x(x-3)}$ in partial fractions.
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Repeated linear factors

- 3 Express $\frac{4x+1}{(x-1)^2}$ in partial fractions of the form $\frac{A}{x-1} + \frac{B}{(x-1)^2}$.
 - 4 Express $\frac{2x^2-3x+1}{x(x-1)^2}$ in partial fractions of the form $\frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$.
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Irreducible quadratic factors

- 5 Express $\frac{3x^2+2}{(x-1)(x^2+1)}$ in partial fractions of the form $\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$.
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Applying partial fractions to integration

- 6 Use partial fractions to evaluate $\int \frac{5x+7}{(x+1)(x+2)} dx$, using the decomposition found earlier in this worksheet.
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Applying partial fractions to a binomial series expansion

- 7 Express $\frac{1}{(1-x)(1-2x)}$ in partial fractions, then use the result to write the first three terms of its series expansion in ascending powers of x .
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Combining terms before decomposing

- 8 Express $\frac{7-x}{(x+1)(x-2)}$ in partial fractions.
 - 9 Express $\frac{x^2+x+1}{x(x+1)}$ in partial fractions. (Hint: the degree of the numerator equals the degree of the denominator, so first perform polynomial division.)
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Applying partial fractions to a rate problem

- 10 A population model gives the rate of change $\frac{dP}{dt} = \frac{1}{P(10-P)}$. Express $\frac{1}{P(10-P)}$ in partial fractions of the form $\frac{A}{P} + \frac{B}{10-P}$.
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Partial fractions with three distinct linear factors

- 11 Express $\frac{6x^2 - 2x - 1}{(x-1)(x+1)(x-2)}$ in partial fractions.
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Partial fractions with an improper fraction and repeated factor

- 12 Express $\frac{x^3}{(x-1)^2}$ in partial fractions, first performing polynomial division since the numerator has higher degree than the denominator.
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Using partial fractions to sum a series

- 13 Express $\frac{1}{n(n+1)}$ in partial fractions, and use the result to find $\sum_{n=1}^4 \frac{1}{n(n+1)}$ by telescoping.
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Decomposing a fraction with a repeated quadratic-adjacent structure

- 14 Express $\frac{x+5}{(x-1)(x+1)^2}$ in partial fractions of the form $\frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$.
- 15 Express $\frac{2x^2 + x - 1}{x^2(x+1)}$ in partial fractions of the form $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}$.
- 16 Use partial fractions to find the Maclaurin (power series) expansion of $\frac{1}{(1-x)(2-x)}$ up to the term in x^2 , first decomposing into the form $\frac{A}{1-x} + \frac{B}{2-x}$.
- 17 Express $\frac{4x^2 - 5x - 15}{(x+1)(x-2)(x+3)}$ in partial fractions, using the cover-up method (evaluating the numerator at each root while covering the matching factor in the denominator).
- 18 A chemical reaction's rate is modeled by $\frac{dC}{dt} = \frac{1}{(C+2)(C-1)}$, where C is a concentration. Express $\frac{1}{(C+2)(C-1)}$ in partial fractions of the form $\frac{A}{C+2} + \frac{B}{C-1}$, and then use the result to write the general antiderivative $\int \frac{1}{(C+2)(C-1)} dC$ in terms of natural logarithms.
- 19 Use the decomposition $\frac{1}{3x} + \frac{8}{3(x-3)}$ found earlier in this worksheet for $\frac{3x-1}{x(x-3)}$ to evaluate $\int_4^5 \frac{3x-1}{x(x-3)} dx$, giving your answer as a single logarithm expression.

- 20 This question has three parts. (a) Factor $x^2 - 4$ and hence express $\frac{x+8}{x^2}$ in partial fractions. (b) Use your result from part (a) to write down $\int \frac{x+8}{x^2-4} dx$. (c) Hence evaluate $\int_3^5 \frac{x+8}{x^2-4} dx$, giving your answer as a single expression involving natural logarithms.